

# MacroPlate



***Your AI-Powered Nutrition Coach & Personal Chef***

## Project Proposal

**Track B Capstone - Mobile Application Development**

**IT 3rd Year | Information Technology**

**Hawassa University - Institute of Technology**

***Submitted to: Elias Jarso***

***eliasjarso@hu.edu.et***

***Submission Date: March 22, 2026***

## Contents

|                                                           |    |
|-----------------------------------------------------------|----|
| The problem? .....                                        | 1  |
| Current challenges include .....                          | 1  |
| Brief User Persona .....                                  | 2  |
| Primary Persona: Biruk .....                              | 2  |
| Secondary Persona: Meron .....                            | 3  |
| The Solution? .....                                       | 3  |
| Key Benefits: .....                                       | 4  |
| Core features (MVP) .....                                 | 4  |
| Feature 1: User Authentication & Onboarding .....         | 4  |
| Feature 2: User Profile & Body Metrics Input .....        | 5  |
| Feature 3: Macro Calculation & Meal Plan Generation ..... | 5  |
| Feature 4: Meal Customization ("Pick Your Protein") ..... | 6  |
| Feature 5: Order & Delivery Scheduling .....              | 6  |
| Future Features (Roadmap) .....                           | 7  |
| Information Architecture .....                            | 7  |
| Technical Stack .....                                     | 10 |
| Project Resources .....                                   | 10 |
| Team Members & Roles .....                                | 11 |

App Name : MacroPlate

Tagline: "Your AI-Powered Nutrition Coach & Personal Chef"

## The problem?

In Ethiopia's rapidly growing fitness community, bodybuilders and gym enthusiasts face a critical gap between training and nutrition. While gyms and coaches are increasingly available across Addis Ababa and regional cities, there is no personalized nutrition guidance tailored to individual body metrics and fitness goals (cutting, maintaining, or bulking).

## Current challenges include

| Challenge                   | Description                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Generic Advice</b>       | Most athletes rely on generic diet charts from the internet that don't account for individual body composition or activity levels  |
| <b>No Macro Tracking</b>    | Calculating daily protein, carb, and fat requirements requires nutrition knowledge that most fitness enthusiasts don't have        |
| <b>Time Constraints</b>     | Even when users know what to eat, meal preparation takes 2-3 hours daily - time that working professionals and students don't have |
| <b>No-Delivery Services</b> | There is no existing service in Ethiopia that bridges the gap between personalized nutrition planning and convenient food delivery |

As a result, dedicated gym-goers spend months training without seeing results and it is not because they train poorly, but because they eat incorrectly. This leads to:

- Wasted gym membership fees
- Demotivation and dropout from fitness

- Potential health risks from improper nutrition

## Brief User Persona

### Primary Persona: Biruk

| Attribute    | Details                                                                                   |
|--------------|-------------------------------------------------------------------------------------------|
| Name         | Biruk                                                                                     |
| Age          | 26                                                                                        |
| Occupation   | Software Engineer                                                                         |
| Location     | Addis Ababa                                                                               |
| Fitness Goal | Bulk (Build muscle)                                                                       |
| Pain Points  | Confused about protein requirements, no time for meal prep, wants science-based nutrition |

Our primary user is Biruk, a university student and gym enthusiast in Addis Ababa. And he says, *“I train 5 days a week but I'm not seeing the muscle growth I expected. I know nutrition is important, but I don't have time to research and prep meals. I need someone to tell me exactly what to eat and bring it to me.”*

## Secondary Persona: Meron

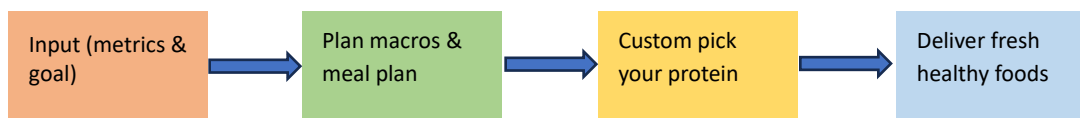
| Attribute           | Details                                                                          |
|---------------------|----------------------------------------------------------------------------------|
| <b>Name</b>         | Meron                                                                            |
| <b>Age</b>          | 24                                                                               |
| <b>Occupation</b>   | Fitness Influencer & Women's Wellness Coach                                      |
| <b>Location</b>     | Addis Ababa                                                                      |
| <b>Fitness Goal</b> | Cut (Weight loss for clients)                                                    |
| <b>Pain Points</b>  | Cannot provide practical meal plans to clients, needs tool to scale her coaching |

Our other user is Meron, struggles to help her clients and she also says, *“My female clients want to lose weight but don't know how to eat. I give them advice, but they struggle to implement it. I need a tool that creates simple, personalized meal plans I can share with them.”*

## The Solution?

MacroPlate is a mobile application that acts as a personalized nutritionist and meal prep service. The user begins by selecting their goal (Cut / Maintain / Bulk) and providing their body metrics (age, weight, height, activity level). Our macroplate IA (intelligent algorithm) calculates daily calorie and macro targets (protein, carbs, fats) and generates a custom weekly meal plan that fits those numbers.

### How it works?



## Key Benefits:

| Benefit              | Description                                                      |
|----------------------|------------------------------------------------------------------|
| <b>Personalized</b>  | Meals based on YOUR body metrics (age, weight, height, activity) |
| <b>Science-Based</b> | Uses BMR formula and proven macro splits                         |
| <b>Flexible</b>      | Swap proteins while maintaining macro targets                    |
| <b>Convenient</b>    | No cooking, no shopping, no planning                             |
| <b>Affordable</b>    | Bulk meal prep pricing, delivered weekly                         |

## Core features (MVP)

### Feature 1: User Authentication & Onboarding

| Sub-feature            | Description                                                             |
|------------------------|-------------------------------------------------------------------------|
| <b>Sign Up</b>         | Email/password registration with OTP verification                       |
| <b>Login</b>           | Email/password and Google sign-in options                               |
| <b>Password Reset</b>  | Forgot password with email reset link                                   |
| <b>Onboarding Flow</b> | Welcome screen, goal selection (Cut/Maintain/Bulk), permission requests |

**Why this is essential:** Users need secure accounts to save their meal plans and orders.

### Feature 2: User Profile & Body Metrics Input

| Sub-feature              | Description                                                 |
|--------------------------|-------------------------------------------------------------|
| <b>Body Metrics</b>      | Age, weight (kg), height (cm), gender                       |
| <b>Activity Level</b>    | 5 options: Sedentary, Light, Moderate, Very Active, Athlete |
| <b>Goal Confirmation</b> | Cut (deficit), Maintain (maintenance), Bulk (surplus)       |

**Why this is essential:** Accurate macro calculation requires accurate input data.

### Feature 3: Macro Calculation & Meal Plan Generation

| Sub-feature             | Description                                                       |
|-------------------------|-------------------------------------------------------------------|
| <b>BMR Calculation</b>  | Harris-Benedict formula for basal metabolic rate                  |
| <b>TDEE Calculation</b> | $BMR \times \text{Activity Factor}$                               |
| <b>Goal Adjustment</b>  | Cut: -500 cal, Maintain: 0 cal, Bulk: +300 cal                    |
| <b>Macro Split</b>      | Protein: 2.0-2.2g per kg, Fats: 25% of calories, Carbs: remainder |
| <b>7-Day Meal Plan</b>  | Breakfast, Lunch, Dinner, Snack for each day                      |

**Why this is essential:** This is the core intelligence of MacroPlate.

#### Feature 4: Meal Customization ("Pick Your Protein")

| Sub-feature                 | Description                                  |
|-----------------------------|----------------------------------------------|
| <b>Customize Button</b>     | Available on every meal card                 |
| <b>Alternative Proteins</b> | Chicken, beef, fish, tofu, eggs with macros  |
| <b>Macro Recalculation</b>  | Daily totals update automatically after swap |
| <b>Price Difference</b>     | Shows extra cost or savings per swap         |

**Why this is essential:** This is MacroPlate's key differentiator - flexibility within structure.

#### Feature 5: Order & Delivery Scheduling

| Sub-feature               | Description                                     |
|---------------------------|-------------------------------------------------|
| <b>Order Summary</b>      | Meal quantities, subtotal, delivery fee, total  |
| <b>Delivery Date</b>      | Calendar picker (minimum tomorrow, max 14 days) |
| <b>Time Slot</b>          | Morning, Afternoon, Evening options             |
| <b>Delivery Address</b>   | Saved addresses + add new address               |
| <b>Payment Method</b>     | Cash on Delivery, Telebirr, Chapa               |
| <b>Order Confirmation</b> | Order number, delivery details, total           |

**Why this is essential:** This completes the loop from planning to delivery.

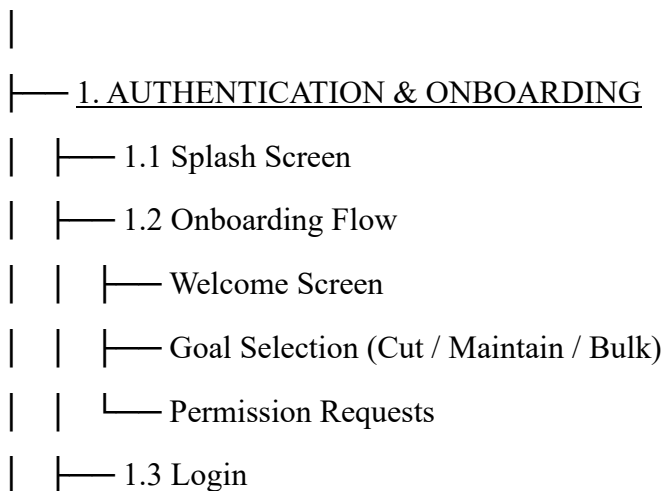


## Future Features (Roadmap)

| Phase   | Feature                  | Description                                            |
|---------|--------------------------|--------------------------------------------------------|
| Phase 2 | Real Payment Integration | Telebirr and Chapa API integration for in-app payments |
| Phase 2 | Meal Logging             | Users log what they ate, see remaining macros          |
| Phase 2 | Progress Tracking        | Weight, measurements, progress photos over time        |
| Phase 3 | Fitness App Integration  | Connect to Google Fit, Fitbit to auto-adjust calories  |
| Phase 3 | Barcode Scanner          | Scan grocery items to see if they fit macro targets    |
| Phase 3 | Social Features          | Share meal plans, compete with friends                 |
| Phase 4 | AI Food Recognition      | Take photo of meal, app estimates macros               |
| Phase 4 | Restaurant Partnerships  | "MacroPlate Approved" meals at local restaurants       |

## Information Architecture

### MACROPLATE - INFORMATION ARCHITECTURE



- | | |— Login with Email
- | | |— Login with Google
- | | |— Forgot Password
- | | |— Sign Up Link
- | |— 1.4 Sign Up
- | |— Sign Up Form
- | |— Email Verification (OTP)
- | |— Profile Setup Redirect

## |— 2. USER PROFILE INPUT

- | |— 2.1 Body Metrics Form
- | | |— Age, Weight, Height, Gender
- | |— 2.2 Activity Level Selection
- | | |— Sedentary, Light, Moderate, Very Active, Athlete
- | |— 2.3 Goal Confirmation
- | | |— Cut, Maintain, Bulk
- | |— 2.4 Submit Button

## |— 3. MAIN DASHBOARD

- | |— 3.1 Header (Greeting, Notifications, Profile)
- | |— 3.2 Daily Summary Card (Calories, Protein, Carbs, Fats)
- | |— 3.3 Quick Actions (View Plan, Order This Week)
- | |— 3.4 Today's Meals (Breakfast, Lunch, Dinner, Snack)

## |— 4. MEAL PLAN MODULE

- | |— 4.1 Weekly View (Day Tabs: Mon - Sun)
- | |— 4.2 Day Cards (Breakfast, Lunch, Dinner, Snack)

- |   └─ 4.3 Meal Detail View (Ingredients, Macros)
- |
- |─ 5. CUSTOMIZATION MODULE ("Pick Your Protein")
- |   └─ 5.1 Current Meal Display
- |   └─ 5.2 Alternative Proteins List (with macros & price)
- |   └─ 5.3 Confirm Changes Button
- |   └─ 5.4 Cancel Button
- |
- |─ 6. ORDER & DELIVERY
- |   └─ 6.1 Order Summary (Quantities, Total)
- |   └─ 6.2 Delivery Scheduling (Date, Time Slot)
- |   └─ 6.3 Delivery Address (Saved, Add New)
- |   └─ 6.4 Payment Method (Cash, Telebirr, Chapa)
- |   └─ 6.5 Place Order Button
- |   └─ 6.6 Order Confirmation
- |
- |─ 7. USER PROFILE
- |   └─ 7.1 Profile Header (Photo, Name, Email)
- |   └─ 7.2 Body Metrics Section (Edit Metrics)
- |   └─ 7.3 Plan History
- |   └─ 7.4 Saved Preferences
- |   └─ 7.5 Account Settings
- |
- |─ 8. NOTIFICATIONS
- |   └─ Order Confirmation, Delivery Reminder
- |   └─ Weekly Plan Ready, Payment Reminder
- |   └─ Meal Prep Reminder

## 9. ERROR STATES

- Network Error (No Internet, Retry)
- Form Validation Errors
- Authentication Errors
- Order Errors

### Technical Stack

| Layer              | Technology                | Purpose                           |
|--------------------|---------------------------|-----------------------------------|
| Frontend           | Android (Kotlin/Java)     | Mobile app development            |
| UI Framework       | Jetpack Compose or XML    | User interface                    |
| Backend            | Firebase / Node.js        | User authentication, data storage |
| Database           | Firebase Firestore / Room | User data, meal plans, orders     |
| Authentication     | Firebase Auth             | Email/Google login                |
| Payments           | Telebirr API, Chapa API   | Payment processing (Phase 2)      |
| Version Control    | Git & GitHub              | Code management                   |
| Project Management | Jira                      | Agile task tracking               |
| Design             | Figma, FigJam             | Wireframes, IA, workflows         |

### Project Resources

| Resource          | Link                                   |
|-------------------|----------------------------------------|
| GitHub Repository | <a href="#">MAD-Project-MacroPlate</a> |
| Figma Wireframes  | <a href="#">View Wireframes</a>        |

| Resource              | Link                        |
|-----------------------|-----------------------------|
| FigJam IA & Workflows | <a href="#">View FigJam</a> |
| Jira Board            | <a href="#">View Jira</a>   |

## Team Members & Roles

- |                               |                                  |
|-------------------------------|----------------------------------|
| 1. UI/UX Implementation Lead  | Helina Betre                     |
| 2. Database & State Lead      | Yonas Atrsie                     |
| 3. Core Logic & Services Lead | Nebiyu Daniel                    |
| 4. Network & API Lead         | Miki Girma                       |
| 5. DevOps & Testing Lead      | Yishak Tadele & Bilisuma Tsegaye |